

DATA ANALYST INTERNSHIP

TASK 6: Sales Trend Analysis Using Aggregations

Objective

Analyze monthly revenue and order volume from the `online_sales` orders table.

Tools

PostgreSQL / MySQL / SQLite

Dataset

Table: `online_sales` (orders table with `order_date`, `amount`, `product_id`).

1. SQL Script

```
-- TASK 6: Sales Trend Analysis Using Aggregations
```

```
-- 1. Monthly revenue and order volume
```

```
SELECT
    EXTRACT(YEAR FROM order_date) AS order_year,
    EXTRACT(MONTH FROM order_date) AS order_month,
    SUM(amount) AS monthly_revenue,
    COUNT(DISTINCT order_id) AS order_volume
FROM online_sales
GROUP BY
    EXTRACT(YEAR FROM order_date),
    EXTRACT(MONTH FROM order_date)
ORDER BY
    order_year,
    order_month;
```

```
-- 2. Monthly sales for a specific time period
```

```
SELECT
    EXTRACT(YEAR FROM order_date) AS order_year,
    EXTRACT(MONTH FROM order_date) AS order_month,
    SUM(amount) AS monthly_revenue,
    COUNT(DISTINCT order_id) AS order_volume
FROM online_sales
WHERE order_date >= '2025-01-01'
    AND order_date < '2026-01-01'
```

```

GROUP BY
    EXTRACT(YEAR FROM order_date),
    EXTRACT(MONTH FROM order_date)
ORDER BY
    order_year,
    order_month;

-- 3. Top 3 months by revenue
SELECT
    EXTRACT(YEAR FROM order_date) AS order_year,
    EXTRACT(MONTH FROM order_date) AS order_month,
    SUM(amount) AS monthly_revenue,
    COUNT(DISTINCT order_id) AS order_volume
FROM online_sales
GROUP BY
    EXTRACT(YEAR FROM order_date),
    EXTRACT(MONTH FROM order_date)
ORDER BY monthly_revenue DESC
LIMIT 3;

```

2. Results Table

The task document specifies a results table as a deliverable, but it does not provide actual order records. Therefore, actual revenue/order values cannot be calculated from the task PDF alone. Run the SQL script on the supplied online_sales dataset and place the returned rows in this table.

Year	Month	Monthly Revenue	Order Volume
YYYY	MM	Calculated by SUM(amount)	Calculated by COUNT(DISTINCT order_id)
YYYY	MM	Calculated by SUM(amount)	Calculated by COUNT(DISTINCT order_id)
YYYY	MM	Calculated by SUM(amount)	Calculated by COUNT(DISTINCT order_id)

3. Procedure / Interpretation

1. Use the order_date column to identify the year and month of each order.
2. Group records by year and month so that each row represents one month.
3. Use SUM(amount) to calculate total monthly revenue.
4. Use COUNT(DISTINCT order_id) to calculate the number of unique orders in each month.
5. Sort the complete trend chronologically using ORDER BY year and month.
6. For a particular period, apply a WHERE condition to restrict the dates before grouping.
7. To identify the highest-sales months, sort monthly revenue in descending order and use LIMIT 3.

4. Expected Outcome

The analysis produces a month-by-month view of revenue and order volume. This makes it possible to observe sales trends over time and identify months with comparatively high or low sales activity.

5. Submission Checklist

- SQL script (.sql)
- Results table (filled using the actual dataset)
- README.md
- Screenshots of query execution/results, if required